

# SOFTWARE ENGINEERING-LAB 01

NAME:DHANUSH S SHEKHAR

SRN:PES1UG24CS662

| Req ID | Type       | Description                                                                                                                                                 | Priority | Acceptance Criteria                                                                                                                                                                                                          | Rationale                                                                              |
|--------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| FR-001 | Functional | The system shall allow creators to submit draft media deliverables for client review and release locked milestone payments upon sponsor approval sign-off.  | High     | <b>Pass:</b> A submitted milestone draft is available for sponsor review, and the corresponding locked milestone payment is released only after sponsor approval. <b>Fail:</b> Payment is released without sponsor approval. | Core escrow workflow connecting deliverable approval with milestone payment release.   |
| FR-002 | Functional | The system shall allow the content creator and client sponsor to define deliverable milestones with milestone descriptions, deadlines, and payment amounts. | High     | <b>Pass:</b> A contract can be saved only when each milestone contains a description, deadline, and payment amount. <b>Fail:</b> A milestone is saved with any required field missing.                                       | Milestones establish the conditions under which deliverables and payments are managed. |

|        |            |                                                                                                                                                    |        |                                                                                                                                                                                                                                  |                                                                                               |
|--------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| FR-003 | Functional | The system shall automatically apply a visible watermark to uploaded image and video draft assets before making them available for sponsor review. | High   | <b>Pass:</b> Every uploaded image/video draft presented to the sponsor contains the configured watermark. <b>Fail:</b> A sponsor can view an unwatermarked draft.                                                                | Protects creator content while allowing sponsors to review draft work.                        |
| FR-004 | Functional | The system shall allow the client sponsor to review watermarked draft deliverables and either approve them or request revisions.                   | High   | <b>Pass:</b> The sponsor can select either "Approve" or "Request Revision," and the system records the selected decision against the milestone. <b>Fail:</b> The system does not record the sponsor's decision.                  | Sponsor approval is the decision point required for milestone completion and payment release. |
| FR-005 | Functional | The system shall display the current status of each milestone and its associated payment to authorized creators and sponsors.                      | Medium | <b>Pass:</b> For every milestone, the system displays a status such as Pending, Submitted, Under Review, Approved, Revision Requested, or Paid. <b>Fail:</b> The displayed status does not match the milestone's recorded state. | Provides transparency about deliverable progress and escrow payment status.                   |

|         |                                |                                                                                                                                                            |      |                                                                                                                                                                                                                                                         |                                                                                                            |
|---------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| NFR-001 | Nonfunctional<br>– Performance | The system shall process automated watermarking for uploaded video/image previews in under 5 seconds.                                                      | High | <b>Pass:</b> At least 95 out of 100 watermarking operations complete within 5 seconds under the specified test load. <b>Fail:</b> More than 5 of 100 operations exceed 5 seconds.                                                                       | Ensures that creators and sponsors do not experience excessive waiting during draft submission and review. |
| NFR-002 | Nonfunctional<br>– Security    | The system shall restrict contract, draft, milestone, and payment information to authenticated and authorized users associated with the relevant contract. | High | <b>Pass:</b> In authorization testing, a user who is not associated with a contract is denied access to that contract's drafts and payment information in 100% of tested unauthorized-access attempts. <b>Fail:</b> Any unauthorized user gains access. | Protects confidential creator content, contract information, and financial information.                    |

## **Actors & Use Cases:**

### **Actor 1 — Content Creator**

The creator:

- creates/defines milestones
- submits draft deliverables
- views milestone status
- receives payment

### **Actor 2 — Client Sponsor**

The sponsor:

- defines/reviews milestones
- reviews drafts
- approves drafts
- requests revisions
- triggers milestone payment release through approval

### **Actor 3 — Payment Gateway**

This is an **external system**, not a human.It:

- processes the milestone payment
- confirms successful payment
- returns payment status

### **UC-01 — Define Contract & Milestones**

Creator and Sponsor define the contract milestones.

### **UC-02 — Submit Draft Deliverable**

Creator uploads the draft.

### **UC-03 — Apply Watermark**

System processes the uploaded media with a watermark.

### **UC-04 — Review Draft**

Sponsor reviews the watermarked draft.

### **UC-05 — Approve Milestone**

Sponsor approves the completed deliverable.

### **UC-06 — Request Revision**

Sponsor requests changes when the draft is not acceptable.

### **UC-07 — Release Milestone Payment**

Payment associated with the approved milestone is released.

### **UC-08 — Process Payment**

Payment Gateway processes the payment transaction.

## **Use-Case Flow:**

UC-02 — Submit Draft Deliverable:

Creator → Upload → Watermark → Sponsor Review

## **Preconditions:**

1. The creator is authenticated.
2. A valid contract exists between the creator and client sponsor.
3. At least one milestone has been defined.
4. The milestone is open for draft submission.
5. The creator has a supported image/video draft available for upload.

## **Postconditions:**

### **Successful case**

1. The draft is uploaded successfully.
2. The system applies the required watermark.
3. The watermarked draft is stored.

4. The draft becomes available to the client sponsor for review.
5. The milestone status changes to **Submitted/Under Review**.

#### **Main Success Scenario:**

1. Creator selects an open milestone and chooses **Submit Draft Deliverable**.
2. System displays the draft upload interface and required milestone information.
3. Creator selects the image/video draft and uploads it.
4. System validates the file type, size, and milestone association.
5. System processes the uploaded media and applies the required watermark.
6. System stores the watermarked draft securely.
7. System marks the milestone as **Submitted/Under Review**.
8. System makes the watermarked draft available to the client sponsor.
9. System notifies the client sponsor that a draft is ready for review.
10. Use case ends successfully.

#### **Alternate Flow 1 — Invalid File**

System detects that the uploaded file has an unsupported format or exceeds the permitted size.

System displays an appropriate error message.

System does not store or submit the invalid file.

Creator is prompted to select another file.

## **Alternate Flow 2-Watermark Processing Failure**

System fails to complete watermark processing.

System does not make the unwatermarked draft available to the sponsor.

System displays a processing failure message.

Creator is asked to retry the upload.